Each counter is a person. To play a circle of 10 people, put counters on spots 1 to 10 of a mat and leave the other spots empty. Start at 1 and go round and round the circle. The people standing take turns: 1 stays standing, the next one sits down, the next one stays standing, the next one sits down, and so on. When you come round past 1 again, keep taking turns from where you left off. Flip a counter over when that person sits down. Skip empty spots and people who are sitting. Stop when only one person is standing.

Problem 1: Play circles of 1 to 16 people on the mat with 16 spots. Write the last one standing for each circle.

people	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
last one standing								

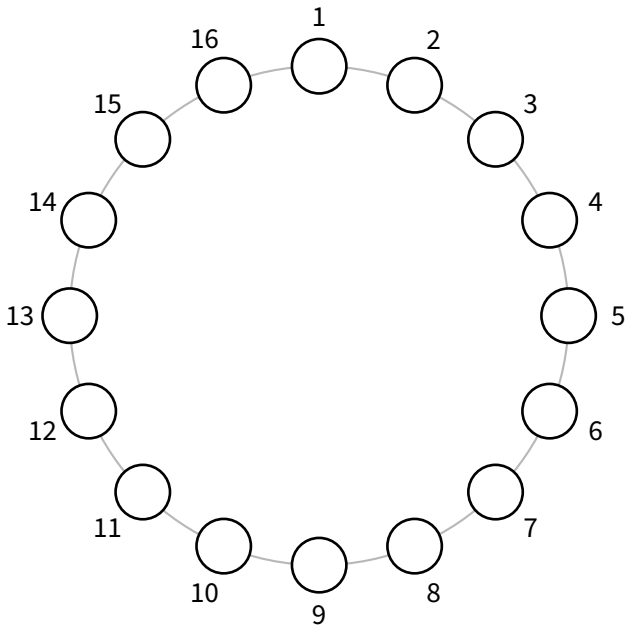
people	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
last one standing								

Problem 2: Predict the last one standing in circles of 20, 25 and 32 people. Then play them on the mat with 32 spots to check.

people	**20**	**25**	**32**
prediction			
last one standing			

Problem 3: Which numbers can never be the last one standing, however many people are in the circle? Explain why.

Problem 4: Explain why number 1 is the last one standing in a circle of 16 people and in a circle of 32 people.



Problem 5: Find every circle up to 40 people where number 9 is the last one standing. Then find every circle up to 40 people where the last one standing has the same number as the number of people in the circle.

Number 9:

Same number:

Problem 6: Who is the last one standing in a circle of 50 people? In a circle of 100 people? Explain how you know.

Problem 7: Now every third person sits down: 1 and 2 stay standing, the next one sits down, the next two stay standing, the next one sits down, and so on. Play circles of 1 to 12 people and write the last one standing for each.

people	**1**	**2**	**3**	**4**	**5**	**6**
last one standing						

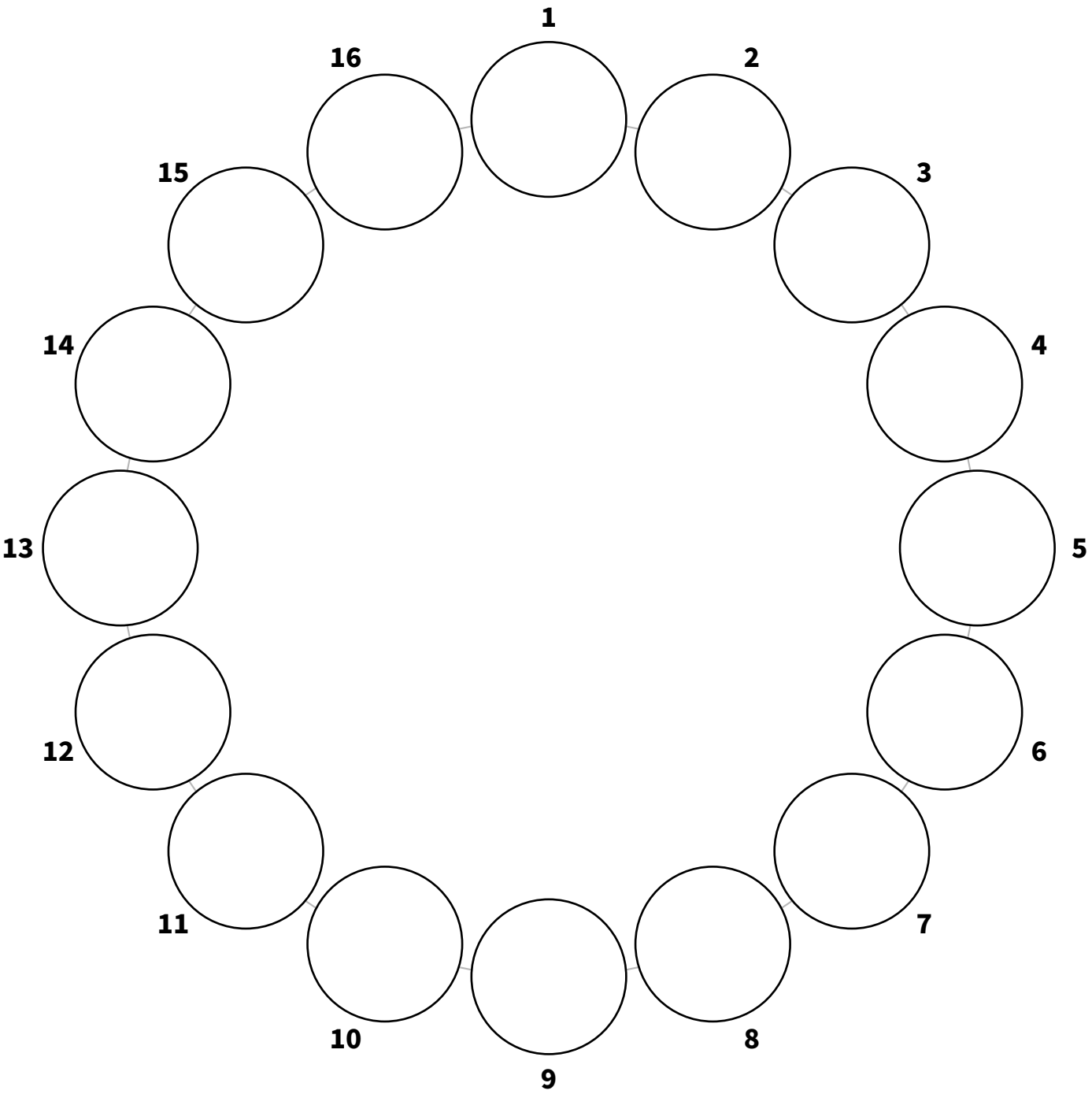
people	**7**	**8**	**9**	**10**	**11**	**12**
last one standing						

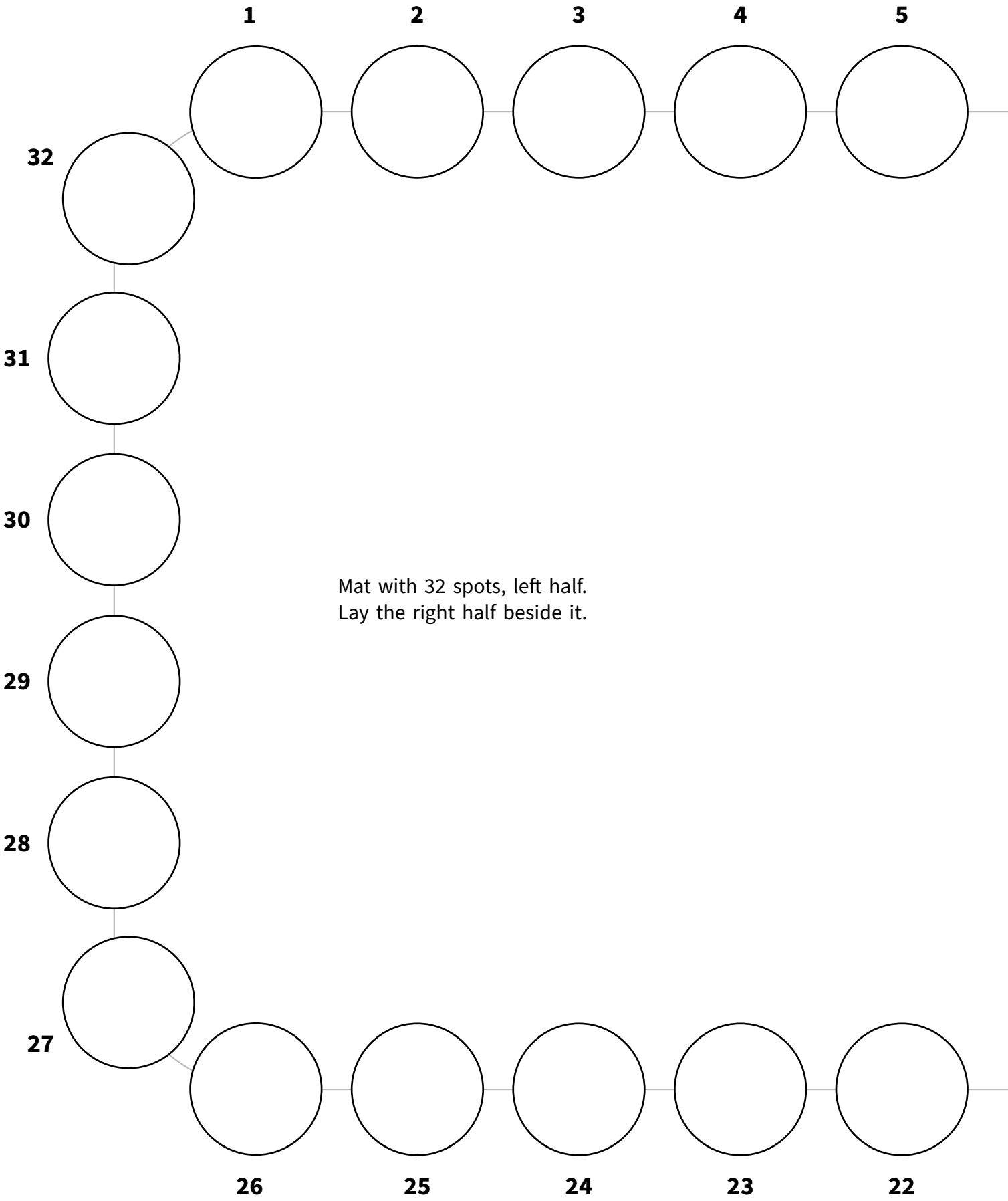
Problem 8: With every third person sitting down, predict the last one standing in circles of 13 to 20 people. Then play circles of 14 and 20 to check.

people	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
prediction								
last one standing								

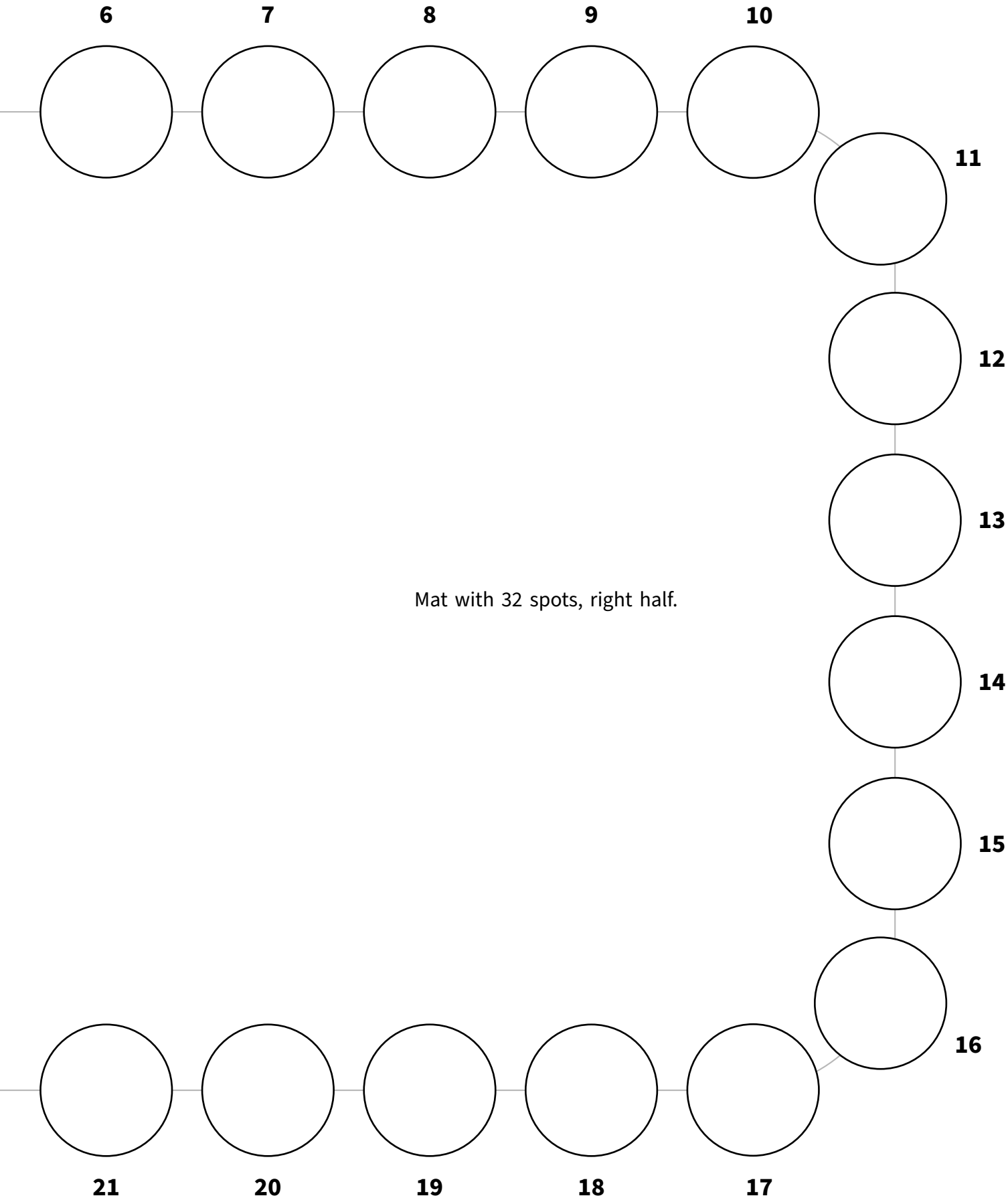
Problem 9: With every third person sitting down, where should you stand to be the last one standing in a circle of 30 people?

Mat with 16 spots.





Mat with 32 spots, left half.
Lay the right half beside it.



Mat with 32 spots, right half.